Pathway attribution with TabDDPM and a meta-learner

1 Pathway

The comparison is not one batch against another. The anchor is a peak day. The next seven days are seven forward steps. On each step a TabDDPM fit only on the peak imputes one feature block, conditional on the other columns of that day's row. The diffusion class is `GaussianMultinomialDiffusion` from `yandex-research/tab-ddpm` (commit `b476257`). Numerical columns use the Gaussian head. Known columns are reset to the anchor noising process at every reverse step, the RePaint conditioning step. The label passed to the denoiser is drawn from the peak-day label distribution. Observed rows are clipped to $[-5, 5]$ before training, and the predicted clean sample is clamped to the same box. On these tables the unclipped reverse chain leaves that box.

Predictive loss is the Brier score of a logistic model fit on the peak and transferred onto day t. On the synthetic panel and MovieLens a ridge on the peak also reports rating or latent MSE. PO-risk follows PermuCATE on rows pooled from the peak ($W = 0$) and day t ($W = 1$): random forests for μ_0, μ_1, and $\hat{\tau}$, logistic propensity with clipping, and

$$\phi(Z) = \frac{(Y - \mu_T)(T - \pi)}{\pi(1 - \pi)} + (\mu_1 - \mu_0), \qquad \text{PO-risk} = \mathbb{E}\big[(\phi - \hat{\tau}(X))^2\big]$$

evaluated on day t with $W = 1$. Replacing block b by its imputed draw changes each loss by Δ_b. The rank share is $|\Delta_b| / \sum_{b'} |\Delta_{b'}|$. MMD compares the block on day t with the same block on the peak. HSIC compares the block with the day's label.

To separate sampler bias from attribution, Table 12 reports a peak-day floor (redraw the block on the anchor) and the excess on the last batch. A verdict of `sampler` means the later batch does not move beyond that redraw; `drift` and `model` require a larger excess, with `drift` also requiring a significant block MMD.

The last day reports a bootstrap percentile interval for each Δ and a 40-permutation p-value for MMD and HSIC. Across the seven steps, a sign-flip test asks whether the block with the largest mean absolute movement exceeds each other block.

2 Synthetic peak

Day 0 is standard normal, with four blocks of four columns: `stable`, `context`, `driver`, and `noise`. The outcome uses `stable` and `driver`. Over the next seven days the `driver` mean moves by up to 1.5 and its outcome coefficient rises, while `context` receives a smaller location shift and no outcome role. The unconditional TabDDPM sample has MMD 0.000 against the peak day.

PO-risk rises from 0.005 to 0.050 and latent MSE from 0.337 to 4.496. By day 7 the `driver` share is 0.905 of the PO-risk movement, 0.959 of the Brier movement, 0.936 of the MSE movement, and 0.834 of the MMD. Its PO interval is $[-0.122, -0.062]$ and its Brier interval is $[-0.139, -0.126]$; both exclude zero. The sign-flip p-values of `driver` against the other three blocks are at most 0.021 for PO-risk, Brier, and MSE. Context has MMD $p = 0.024$, and its Brier movement stays near 0.001. HSIC deltas are all below 0.001 and do not rank the blocks. Table 1–3 record the path, the peak day, and the sign-flip tests.

3 IEEE-CIS week after the fraud-rate peak

The local fraud-rate peak with seven usable days after it is transaction day 173 (fraud rate 0.0495). The pathway runs through day 180 (0.0374). Six blocks are used: amount, card, address, counts, timedelta, and hour. The monotone clock is not a feature. The unconditional sample has MMD 0.002 against the peak day.

No block has an MMD p-value below 0.05 on day 180. The largest mean absolute PO-risk movement is timedelta, and its sign-flip p-values against card, address, and counts are all above 0.178. Every day-180 PO interval includes zero. The transferred Brier movement is largest for card. Its sign-flip p-value is 0.008 against amount and address and 0.028 against counts; it is 0.054 against hour and 0.292 against timedelta. Card's day-180 Brier interval is $[-0.023, -0.004]$, and its MMD p-value is 1. Over this week the card block moves the fraud model's predictions, and the covariate blocks do not separate. Tables 4 and 5 give the day-180 numbers and the seven-step tests.

4 MovieLens week after the high-rating peak

The local peak in the rate of ratings of at least 4, among days that still have seven later calendar days, is 2001-01-01 (rate 0.615). The pathway runs through 2001-01-08 (0.483). Rating is the outcome. The user block is gender, age, occupation, and the first zip digit. The item block is eight genre indicators. Context is the hour; weekday is constant inside one calendar day and drops out. The peak day has 633 rows after keeping users and items with at least 20 ratings. The unconditional sample has MMD 0.004 against that day.

Mean MMD over the seven days is 0.108 for hour, 0.100 for user, and 0.012 for genre. Hour against genre has sign-flip $p = 0.016$; hour against user has $p = 0.413$. On the last day every block's MMD p-value sits at the permutation floor 0.024, while the genre MMD itself is 0.006.

Predictive loss points at genre. Mean absolute Brier movement is 0.010 for genre, 0.005 for hour, and 0.004 for user. The sign-flip p-value of genre against user is 0.029, and against hour 0.074. On 2001-01-08 the genre Brier interval is $[-0.025, -0.010]$ and the user Brier interval includes zero. Rating MSE also leads on genre (0.044 against 0.027 for user), with sign-flip $p = 0.125$ against user. Mean absolute PO-risk movement is 0.009 for user and 0.005 for genre, with sign-flip $p = 0.198$. The last-day user PO interval is $[-0.048, -0.008]$; the genre PO interval includes zero. HSIC deltas stay at most 0.002.

The covariate shift over this week is who is rating and at which hour. The genre margin barely moves, and Brier still ranks genre first because the peak-day rating model uses genre. Tables 6–8 record the path.

5 Home Credit along application id

`application_train.csv` has no event time. Rows sorted by `SK_ID_CURR` are cut into 40 equal slices. The local peak in the default rate is slice 31 (0.083), and the pathway runs to slice 38 (0.080). The five blocks are external (the three `EXT_SOURCE` columns), amount, personal, region, and other. Each slice contributes 700 rows. The unconditional sample has MMD 0.003 against the peak slice.

Every block MMD stays below 0.005. The largest seven-step MMD is external, and its sign-flip p-value against each other block is at least 0.266. On the last slice only `other` has MMD $p = 0.049$, on a statistic of 0.004.

Brier isolates external. Its mean absolute movement is 0.023, and the sign-flip p-value against each of the other four blocks is 0.008. The last-slice Brier interval is $[-0.044, -0.006]$. PO-risk does not separate external from amount (mean absolute movement 0.005 and 0.004, sign-flip $p = 0.253$). On the

Table 1: Synthetic pathway. The anchor is day 0. Each step is one day closer to the drift peak. Shares are the block's fraction of the total absolute movement.

step	PO-risk	Brier	MSE	driver PO	driver Brier	driver MSE	driver MMD	context MMD
1	0.656	0.236	0.337	0.447	0.420	0.492	0.863	0.000
2	0.720	0.231	0.403	0.764	0.672	0.548	0.886	0.076
3	0.763	0.212	0.486	0.946	0.650	0.689	0.905	0.095
4	0.918	0.195	0.901	0.983	0.868	0.810	0.920	0.071
5	1.322	0.182	1.427	0.985	0.822	0.826	0.864	0.136
6	0.863	0.158	2.589	0.989	0.911	0.921	0.852	0.144
7	1.176	0.122	4.496	0.966	0.959	0.936	0.834	0.166

Table 2: Synthetic day 7, the drift peak. Deltas are the change in the loss when that block is replaced by the TabDDPM draw from the anchor conditional. Intervals are bootstrap percentiles. MMD and HSIC p-values use 40 permutations.

block	PO Δ	PO interval	Brier Δ	Brier interval	MSE Δ	MMD	MMD p	HSIC Δ	HSIC p	PO share	Brier share
stable	-1.645	[-4.537, -0.191]	-0.004	[-0.008, 0.001]	-0.292	0.000	1.000	-0.0004	0.585	0.003	0.028
context	-15.743	[-25.812, -3.465]	-0.001	[-0.002, -0.000]	-0.035	0.084	0.024	-0.0002	0.341	0.029	0.010
driver	-531.357	[-879.121, -224.706]	-0.132	[-0.139, -0.126]	-4.835	0.420	0.024	0.0003	0.049	0.966	0.959
noise	-1.374	[-4.484, -0.227]	-0.000	[-0.002, 0.001]	0.001	0.000	1.000	-0.0002	0.024	0.002	0.003

last slice those two PO intervals exclude zero, and the personal, region, and other intervals include zero. HSIC deltas are all below 0.001.

Along application id the covariates do not separate. The peak-day default model uses the external scores, so replacing that block moves the Brier score transferred onto the later slices. Tables 9–11 record the path.

Table 3: Sign-flip test over the seven synthetic steps. The contrast is the absolute movement of the driver minus the other block.

loss	comparison	mean gap	p
PO-risk	driver > stable	241.247	0.008
PO-risk	driver > context	238.468	0.008
PO-risk	driver > noise	241.608	0.008
Brier	driver > stable	0.050	0.014
Brier	driver > context	0.060	0.008
Brier	driver > noise	0.060	0.008
MSE	driver > stable	1.511	0.014
MSE	driver > context	1.821	0.008
MSE	driver > noise	1.828	0.008

Table 4: IEEE-CIS transaction day 180, seven days after the fraud-rate peak on day 173. Same columns as the synthetic peak day. No latent MSE on this table.

block	PO Δ	PO interval	Brier Δ	Brier interval	MMD	MMD p	HSIC Δ	HSIC p	PO share	Brier share
amount	-0.028	[-0.136, -0.012]	-0.000	[-0.000, 0.000]	0.002	0.098	-0.0001	0.122	0.104	0.002
card	-0.049	[-0.145, -0.021]	-0.013	[-0.023, -0.004]	0.001	1.000	-0.0003	0.098	0.185	0.420
address	-0.046	[-0.089, -0.009]	0.005	[0.002, 0.008]	0.000	1.000	0.0000	0.073	0.174	0.159
counts	-0.103	[-0.308, -0.036]	-0.003	[-0.006, 0.002]	0.000	0.146	-0.0000	0.122	0.385	0.102
timedelta	-0.027	[-0.065, -0.009]	0.009	[-0.003, 0.023]	0.000	1.000	-0.0001	0.049	0.102	0.302
hour	-0.014	[-0.059, -0.003]	-0.000	[-0.005, 0.004]	0.002	0.073	0.0002	0.024	0.051	0.015

Table 5: Sign-flip test over the seven IEEE-CIS steps after the peak. PO-risk does not name a single block. The transferred Brier movement is largest for the card block.

loss	comparison	mean gap	p
PO-risk	counts > amount	0.199	0.008
PO-risk	counts > card	0.174	0.008
PO-risk	counts > address	0.211	0.008
PO-risk	counts > timedelta	0.210	0.008
PO-risk	counts > hour	0.220	0.008
Brier	card > amount	0.008	0.008
Brier	card > address	0.006	0.008
Brier	card > counts	0.005	0.028
Brier	card > timedelta	0.002	0.292
Brier	card > hour	0.004	0.054

Table 6: MovieLens-1M, seven calendar days after a peak in the high-rating rate. Rating is the outcome. Shares use absolute movement.

step	rate	PO-risk	Brier	PO top	PO share	Brier top	Brier share	MSE top	MSE share	MMD top	MMD share
2001-01-02	0.503	2.569	0.263	item	0.605	context	0.692	user	0.769	user	0.825
2001-01-03	0.597	0.627	0.255	user	0.734	item	0.737	user	0.474	user	0.655
2001-01-04	0.500	2.650	0.244	item	0.584	context	0.426	item	0.601	context	0.698
2001-01-05	0.539	0.794	0.255	user	0.437	item	0.628	item	0.708	user	0.670
2001-01-06	0.657	0.625	0.251	user	0.894	item	0.617	item	0.599	context	0.498
2001-01-07	0.551	1.204	0.251	user	0.643	item	0.695	item	0.508	context	0.618
2001-01-08	0.490	2.935	0.246	user	0.524	item	0.593	item	0.497	context	0.582

Table 7: MovieLens last day of the pathway. Rating MSE is the predictive companion of Brier.

block	PO Δ	PO interval	Brier Δ	Brier interval	MSE Δ	MMD	MMD p	HSIC Δ	HSIC p	PO share	Brier share
user	-3.461	[-10.795, -2.257]	-0.008	[-0.014, 0.001]	-0.047	0.052	0.024	0.0021	0.024	0.524	0.284
item	-1.975	[-5.573, -0.715]	-0.018	[-0.025, -0.010]	-0.048	0.006	0.024	-0.0000	0.024	0.299	0.593
context	-1.167	[-7.280, -0.291]	0.004	[0.001, 0.006]	-0.002	0.081	0.024	0.0019	0.024	0.177	0.123

Table 8: Sign-flip test across the seven steps. The contrast is absolute movement.

loss	comparison	mean gap	p
PO-risk	user > item	0.538	0.400
PO-risk	user > context	4.398	0.023
Brier	item > user	0.005	0.029
Brier	item > context	0.005	0.074
MSE	item > user	0.017	0.125
MSE	item > context	0.030	0.016

Table 9: Home Credit along SK_ID_CURR. Seven slices after a peak in the default rate. There is no event time. Every block MMD stays below 0.005. An MMD entry of – is a total below 10^{-4}, and a large MMD share is still a split of a near-zero total.

step	rate	PO-risk	Brier	PO top	PO share	Brier top	Brier share	MMD top	MMD share
slice 32	0.079	0.183	0.202	external	0.411	external	0.695	external	0.925
slice 33	0.097	0.194	0.201	external	0.478	external	0.500	other	0.531
slice 34	0.067	0.149	0.210	external	0.460	external	0.437	amount	0.423
slice 35	0.084	0.203	0.193	external	0.439	external	0.702	external	0.939
slice 36	0.090	0.207	0.213	external	0.493	external	0.379	–	–
slice 37	0.086	0.215	0.201	external	0.348	external	0.366	–	–
slice 38	0.074	0.177	0.207	external	0.385	external	0.508	–	–

Table 10: Home Credit last slice. PO-risk and transferred Brier on the default label.

block	PO Δ	PO interval	Brier Δ	Brier interval	MMD	MMD p	HSIC Δ	HSIC p	PO share	Brier share
external	-0.066	[-0.142, -0.042]	-0.026	[-0.044, -0.006]	0.000	0.317	-0.0005	0.024	0.385	0.508
amount	-0.015	[-0.049, -0.011]	-0.003	[-0.006, 0.002]	0.000	0.317	-0.0001	0.512	0.088	0.058
personal	-0.010	[-0.039, -0.005]	0.004	[-0.002, 0.010]	0.000	1.000	-0.0000	0.195	0.059	0.076
region	-0.054	[-0.151, -0.039]	0.015	[0.010, 0.024]	0.000	0.195	-0.0002	0.390	0.315	0.284
other	-0.026	[-0.062, -0.017]	0.004	[-0.006, 0.010]	0.000	0.049	0.0001	0.073	0.153	0.074

Table 11: Sign-flip test across the seven steps. The contrast is absolute movement.

loss	comparison	mean gap	p
PO-risk	external > amount	0.070	0.008
PO-risk	external > personal	0.066	0.008
PO-risk	external > region	0.043	0.008
PO-risk	external > other	0.068	0.008
Brier	external > amount	0.020	0.008
Brier	external > personal	0.018	0.008
Brier	external > region	0.015	0.008
Brier	external > other	0.018	0.008

Table 12: Sampler floor against the last batch. The floor is the Brier change from redrawing that block on the peak day. Excess is the last-batch Brier change minus the floor. Shares use absolute values. The anchor imputation MMD does not use a later batch. A verdict of sampler means the excess is smaller than the floor. Drift means the excess is at least twice the floor and the block MMD has $p < 0.05$. Model means the same excess bar without that MMD. The factor of two is a reporting bar. Thin means the excess beats the floor but not by two. None means both pieces are under 10^{-4}.

data	block	floor	day	excess	raw share	excess share	anchor impute MMD	day MMD	clears p	verdict
synthetic	stable	-0.011	-0.004	0.007	0.028	0.052	0.001	0.000	0.489	sampler
synthetic	context	0.000	-0.001	-0.002	0.010	0.012	0.000	0.084	0.016	drift
synthetic	driver	-0.012	-0.132	-0.120	0.959	0.914	0.000	0.420	0.008	drift
synthetic	noise	-0.003	-0.000	0.003	0.003	0.023	0.002	0.000	1.000	sampler
IEEE	amount	-0.000	-0.000	-0.000	0.002	0.001	0.021	0.002	0.021	none
IEEE	card	-0.008	-0.013	-0.005	0.420	0.148	0.002	0.001	0.416	sampler
IEEE	address	0.000	0.005	0.005	0.159	0.123	0.012	0.000	0.008	model
IEEE	counts	0.002	-0.003	-0.005	0.102	0.144	0.101	0.000	0.295	model
IEEE	timedelta	-0.010	0.009	0.020	0.302	0.532	0.001	0.000	0.955	thin
IEEE	hour	0.001	-0.000	-0.002	0.015	0.052	0.039	0.002	0.059	thin
MovieLens	user	-0.013	-0.008	0.005	0.284	0.362	0.008	0.052	1.000	sampler
MovieLens	item	-0.021	-0.018	0.003	0.593	0.270	0.001	0.006	1.000	sampler
MovieLens	context	-0.001	0.004	0.005	0.123	0.368	0.007	0.081	0.008	drift
Home Credit	external	-0.019	-0.026	-0.008	0.508	0.172	0.005	0.000	0.141	sampler
Home Credit	amount	-0.002	-0.003	-0.001	0.058	0.019	0.005	0.000	0.174	sampler
Home Credit	personal	-0.006	0.004	0.010	0.076	0.232	0.009	0.000	0.845	thin
Home Credit	region	0.002	0.015	0.013	0.284	0.288	0.003	0.000	0.016	model
Home Credit	other	-0.009	0.004	0.013	0.074	0.289	0.010	0.000	1.000	thin